

Section - A

Q: 1(a) $(1, 1)$ and $(1, a)$ are L.I.
if $\begin{vmatrix} 1 & 1 \\ 1 & a \end{vmatrix} = 0 \Rightarrow a = 1$ — (1)

Q: 1(b) $(1, -2, k)$ is L.C. of
 $(3, 0, -2), (2, -1, -5)$ if

$$\begin{vmatrix} 3 & 0 & -2 \\ 2 & -1 & -5 \\ 1 & -2 & k \end{vmatrix} = 0 \Rightarrow k = -8$$
 — (0.5)

Q: 1(c) Null Space:

$$T(x_1, x_2) = (0, 0)$$

$$\Rightarrow (x_1 + x_2, x_1) = (0, 0)$$
 — (0.5)

$$\Rightarrow x_1 + x_2 = 0 \text{ \& } x_1 = 0 \Rightarrow x_2 = 0$$

$$\Rightarrow N(T) = \{(0, 0)\} \Rightarrow n(T) = 0$$
 — (0.5)

Q: 1(d) Let V be a Vector Space
over a field F . Let B be a
subset of $V(F)$ then, B is a
basis of $V(F)$ if

(i) elements of B are L.I.

(ii) B spans $V(F)$. — (1)

Q: 1(e) $\begin{vmatrix} 0-\lambda & -1 \\ 2 & 3-\lambda \end{vmatrix} = 0 \Rightarrow \lambda^2 - 3\lambda + 2 = 0$
 $\Rightarrow \lambda = 1, 2$ — (1)

Q: 1(f) $(iA)^\theta = -(iA)$

$$-i(A^\theta) = -iA$$

$$\Rightarrow A^\theta = A$$

$\Rightarrow A$ is Hermitian matrix. — (1)

Section - B

Q: 2(a) B is a basis of $\mathbb{R}^3$

if (i) Elements of B are L.I.

(ii) $\dim(B) = \dim(\mathbb{R}^3) = 3$

(i) Elements of B are L.I.:

$$\begin{vmatrix} 2 & 1 & 4 \\ 1 & -1 & 2 \\ 3 & 1 & -2 \end{vmatrix} = 2(2-2) - 1(-2-6) + 4(1+3) = 20 \neq 0$$

$\Rightarrow$ Elements of B are L.I. — (2)

$\Rightarrow \dim(B) = 3 = \dim(\mathbb{R}^3)$

$\Rightarrow B$ is a basis of $\mathbb{R}^3$. — (1)

Q: 2(b) $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 3 \\ 2 & 1 & 0 \\ -3 & 1 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 3 \\ 0 & -1 & -2 \\ 0 & 4 & 5 \end{bmatrix}$

$$\sim \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \\ 0 & 0 & -7 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$\Rightarrow$ Basis of W

$$= \{(1, 1, 1), (0, 1, 3), (0, 0, 1)\}$$
 — (2)

dimension of $W = 3$ — (1)

Section - C

3(a) $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$

$$T(x, y, z) = (2x - 3y, 7y + 2z)$$

Let $(x_1, y_1, z_1), (x_2, y_2, z_2) \in \mathbb{R}^3$

(i) $T((x_1, y_1, z_1) + (x_2, y_2, z_2))$

$$= T(x_1 + x_2, y_1 + y_2, z_1 + z_2)$$

$$= (2(x_1 + x_2) - 3(y_1 + y_2),$$

$$7(y_1 + y_2) + 2(z_1 + z_2))$$

$$T((x_1, y_1, z_1) + (x_2, y_2, z_2))$$

$$= (2x_1 - 3y_1, 7y_1 + 2z_1)$$

$$+ (2x_2 - 3y_2, 7y_2 + 2z_2)$$

$$= T(x_1, y_1, z_1) + T(x_2, y_2, z_2) \quad \text{--- (2)}$$

(ii) let c be any scalar,
then

$$T(c(x, y, z)) = T(cx, cy, cz)$$

$$= (2cx - 3cy, 7cy + 2cz)$$

$$= c(2x - 3y, 7y + 2z)$$

$$= cT(x, y, z) \quad \text{--- (1)}$$

$\Rightarrow T$ is a linear transformation.

3(b) Two basis of R^2 :

$$S = \{u_1, u_2\} = \{(1, 2), (3, 5)\}$$

$$S' = \{v_1, v_2\} = \{(1, -1), (1, -2)\}$$

Change of basis matrix from
 S to S' .

$$\rightarrow (x, y) = au_1 + bu_2$$

$$\text{let } (x, y) = a(1, 2) + b(3, 5)$$

$$a + 3b = x$$

$$2a + 5b = y$$

$$\Rightarrow a = 3y - 5x, b = 2x - y \quad \text{--- (1)}$$

$$\therefore (x, y) = (3y - 5x)(1, 2)$$

$$+ (2x - y)(3, 5)$$

$$v_1 = (1, -1) = -8u_1 + 3u_2$$

$$v_2 = (1, -2) = -11u_1 + 4u_2 \quad \text{--- (1)}$$

(2)

$\therefore$ change of basis matrix

$$P = \begin{bmatrix} -8 & -11 \\ 3 & 4 \end{bmatrix} \quad \text{--- (1)}$$

$$4(a) A = \begin{bmatrix} 2 & -1 & 3 & 1 \\ 1 & 4 & -2 & 1 \\ 5 & 2 & 4 & 3 \end{bmatrix}$$

$$R_1 \leftrightarrow R_2$$

$$\sim \begin{bmatrix} 1 & 4 & -2 & 1 \\ 2 & -1 & 3 & 1 \\ 5 & 2 & 4 & 3 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 4 & -2 & 1 \\ 0 & -9 & 7 & -1 \\ 0 & -18 & 14 & -2 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - 2R_2$$

$$\sim \begin{bmatrix} 1 & 4 & -2 & 1 \\ 0 & -9 & 7 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$R_2 \rightarrow \frac{R_2}{-9}$$

$$\sim \begin{bmatrix} 1 & 4 & -2 & 1 \\ 0 & 1 & -\frac{7}{9} & \frac{1}{9} \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

which is Echelon form.

rank of matrix

= no. of non-zero rows

$$= 2$$

--- (3)

(3)

$$\begin{aligned}
 4(b) \quad x_1 + 2x_2 + x_3 &= 2 \\
 3x_1 + x_2 - 2x_3 &= 1 \\
 4x_1 - 3x_2 - x_3 &= 3 \\
 2x_1 + 4x_2 + 2x_3 &= 4
 \end{aligned}$$

Augmented matrix

$$\begin{aligned}
 C &= [A \cdot B] \\
 &= \begin{bmatrix} 1 & 2 & 1 & 2 \\ 3 & 1 & -2 & 1 \\ 4 & -3 & -1 & 3 \\ 2 & 4 & 2 & 4 \end{bmatrix}
 \end{aligned}$$

$$R_2 \rightarrow R_2 - 3R_1, R_3 \rightarrow R_3 - 4R_1,$$

$$R_4 \rightarrow R_4 - 2R_1$$

$$C \sim \begin{bmatrix} 1 & 2 & 1 & 2 \\ 0 & -5 & -5 & -5 \\ 0 & -11 & -5 & -5 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$R_2 \rightarrow \frac{R_2}{-5}, R_3 \rightarrow -R_3$$

$$C \sim \begin{bmatrix} 1 & 2 & 1 & 2 \\ 0 & 1 & 1 & 1 \\ 0 & 11 & 5 & 5 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - 11R_2$$

$$C \sim \begin{bmatrix} 1 & 2 & 1 & 2 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & -6 & -6 \end{bmatrix}$$

$$\rho(A) = \rho(C) = 3$$

= no. of unknowns

$\Rightarrow$ unique solution exists. ⁽²⁾

Equivalent system of equation is

$$x_1 + 2x_2 + x_3 = 2$$

$$x_2 + x_3 = 1$$

$$-6x_3 = -6$$

$$\Rightarrow x_3 = 1$$

$$\Rightarrow x_1 = 1, x_2 = 0, x_3 = 1.$$

— X — ⁽¹⁾